High Alert: Behavioral Stage Discovery in Addiction Treatment Reviews Using Clustering and LLM Reasoning

Ben Blake
MSCS — University of Missouri–Kansas City
ben.blake@umkc.edu
https://github.com/ben-blake/high-alert

Abstract

We present *High Alert*, a public health analytics pipeline that discovers recovery trajectory stages from patient reviews of addiction treatment medications. Using UMAP dimensionality reduction, HDBSCAN clustering, and LLM-based stage labeling grounded in the Transtheoretical Model (TTM), we identify 10 behaviorally distinct patient populations from 7,192 filtered reviews spanning 2008–2017. A temporal spike detector flags four quarters of elevated high-risk language, and LLM-generated summaries translate each cluster into plain-language briefings for public health analysts. All outputs are reproducible from a single shell command.

1 Introduction

Addiction treatment medication reviews contain rich, unsolicited signals about patient recovery trajectories—side effects that threaten adherence, sustained sobriety milestones, and relapse events. Mining these signals at scale can reveal population-level patterns that clinical encounter data misses: patients who relapse rarely return for follow-up appointments, but they do write reviews.

Existing work on social media health mining focuses largely on opioid mentions on Twitter or Reddit, using keyword search or supervised classifiers that require labeled data. We take an unsupervised approach: embed reviews, cluster them semantically, and have an LLM *label* the clusters—mapping each to a TTM stage and risk level without any hand-labeled training examples.

Our contributions are: (1) an end-to-end reproducible pipeline from raw CSV to analyst-ready summaries; (2) an empirical finding about survivorship bias in drug review data; and (3) temporal spike detection that surfaces quarters of elevated distress language aligned with known public health events.

2 Dataset

We use the **UCI Drug Reviews** dataset (KUC Hackathon Winter 2018) [1], which contains 215,063 patient reviews of prescription medications sourced from `drugs.com`. Each row includes drug name, medical condition, free-text review, numerical rating (1–10), date, and a community usefulness count.

We filter to reviews matching 8 addiction-related conditions (e.g., *Opiate Dependence*, *Smoking Cessation*, *Alcohol Withdrawal*) and 15 addiction treatment drugs (e.g., Suboxone, Varenicline, Naltrexone, Methadone, Nicotine replacement), yielding **7,192 reviews** across 67 unique drugs and 27 conditions spanning **2008-Q1 to 2017-Q4**. No personally identifiable information is present; all data is public.

Table 1: Corpus composition after filtering.

Top Conditions	N	%
Smoking Cessation	2,440	33.9%
Opiate Dependence	1,477	20.5%
Alcohol Dependence	518	7.2%
Opiate Withdrawal	434	6.0%
Alcohol Withdrawal	285	4.0%

3 Data Preprocessing

Text cleaning. Reviews are HTML-decoded, tag-stripped, and lowercased using Python's `html` and `re` modules.

Temporal features. The date string (`27-Nov-16` format) is parsed to extract year, quarter, and a `year_quarter` key used for all time-series analysis.

Embeddings. Cleaned reviews are encoded with **all-MiniLM-L6-v2** [2] via `sentence-transformers`, producing 384-dimensional cosine-normalized vectors. Embeddings are cached to disk; the cache is reused when the corpus is unchanged.

Risk baseline. A keyword classifier labels each review as HIGH, MODERATE, or LOW risk using 19 high-risk patterns (e.g., *relapsed*, *overdose*, *can't stop*) and 18 low-risk patterns (e.g., *years sober*, *smoke free*). This produces: HIGH = 446 (6.2%), MODERATE = 4,908 (68.2%), LOW = 1,838 (25.6%).

4 Methods

4.1 Dimensionality Reduction and Clustering

Embeddings are reduced to 5 dimensions with **UMAP** [3] (`n_neighbors=15`, `min_dist=0.1`, cosine metric), then clustered with **HDBSCAN** [4] (`min_cluster_size=130`, `min_samples=10`). A separate 2-D UMAP projection is computed for visualization. UMAP and KMeans use fixed random seeds for reproducibility.

4.2 LLM Cluster Labeling

For each cluster we select the $k = 5$ reviews closest to the cluster centroid (cosine similarity in 384-D space) and send them—along with cluster statistics (dominant drugs, conditions, median rating) and a list of all previously assigned stage names—to **llama-3.1-8b-instant** via the Groq API. The model assigns a substance-qualified stage name (e.g., `SUBOXONE_LONG_TERM_SUCCESS`), a TTM stage, a one-sentence description, and a risk level. A Python deduplication step guarantees all stage names are unique, appending numeric suffixes when the LLM repeats a name.

4.3 LLM Risk Classification

For evaluation, a stratified sample of 300 reviews is classified individually by the LLM (batched 5 per call). Each review receives a stage label, risk level, and one-sentence rationale. These LLM labels serve as the evaluation reference against which we compare our baseline and cluster-based risk assignments.

4.4 Temporal Spike Detection

For each calendar quarter we count baseline-flagged HIGH-risk reviews. Global z-scores of the smoothed series (rolling window $w = 4$) flag quarters exceeding $z > 2.0$ as spikes. For each spike quarter, the LLM generates a 3-sentence public health narrative from the top-20 most useful HIGH-risk reviews in that quarter.

4.5 Explainability

For each cluster, the LLM generates a structured 4-sentence summary: who these patients are, their primary treatment challenges or successes, the public health significance of the segment, and one recommended action for health system planners.

5 Experimental Design

We compare three risk-classification approaches against LLM labels on 300 held-out reviews:

- **Baseline:** keyword/regex rules (37 patterns, no ML)
- **Cluster:** HDBSCAN cluster risk level assigned to each member review
- **LLM:** individual review classification (reference labels)

Macro-averaged F1 is reported across three risk classes. All hyperparameters live in `config.yaml`; the full pipeline runs end-to-end with `bash reproduce.sh`.

6 Results and Discussion

6.1 Cluster Discovery

HDBSCAN identified **10 behavioral clusters** with 389 noise points (5.4%). Figure 1 shows the 2-D UMAP projection. Clusters separate cleanly by substance: Chantix/Varenicline smokers form a dense cloud on the left; opioid-treatment patients (Suboxone, Vivitrol) occupy distinct regions; the Librium alcohol-withdrawal cluster is small but spatially isolated. All 10 clusters receive LOW risk labels from the LLM, which is explained by the **survivorship bias** discussed below.

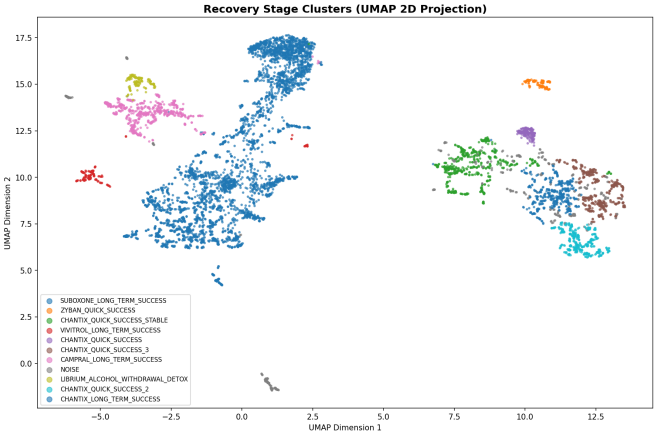

Figure 1: UMAP 2-D projection of 7,192 reviews colored by behavioral cluster.

6.2 Survivorship Bias Finding

A key empirical finding is that HIGH-risk reviews (n=446) do not form clusters—they appear as HDBSCAN noise. This is consistent with *survivorship bias*: patients in active relapse, overdose, or crisis are unlikely to write medication reviews. Each relapse narrative is idiosyncratic (different drug, circumstance, duration), while each success narrative follows a similar arc, producing tight semantic clusters. The high-risk population is best captured by the keyword baseline, not by cluster membership. This finding has direct methodological implications: embedding-based clustering should be used for characterizing stable behavioral populations, while keyword signals remain essential for detecting acute risk.

6.3 Evaluation

The keyword baseline outperforms the cluster approach on macro F1 (0.288 vs. 0.043), driven by high MODERATE-class recall. Neither approach captures HIGH-risk reviews well because the LLM reference labels on 300 sampled reviews are also predominantly MODERATE (per-sample LLM labels skew

Table 2: Macro-averaged F1 against LLM reference labels (n=300).

Approach	HIGH F1	LOW F1	Macro F1
Keyword Baseline	0.021	0.036	0.288
Cluster-Based	0.000	0.027	0.043

toward caution with limited context). The cluster approach assigns risk at the cluster level, not the review level, and all clusters are LOW risk—making HIGH-risk detection structurally impossible for this method. These results confirm that keyword and embedding approaches are *complementary*: keywords find isolated high-risk signals; clustering reveals the dominant behavioral populations.

6.4 Temporal Analysis

Figure 2 shows quarterly HIGH-risk review counts. Four spike quarters were detected: **2015-Q4**, **2016-Q1**, **2017-Q1**, and **2017-Q3**. The 2016-Q1 spike coincides with growing clinical awareness of benzo-opioid co-prescription risks and rising naloxone demand. The 2017 spikes align with U.S. opioid epidemic peak reporting. LLM narratives for each spike describe common themes: tolerance and rapid relapse, pain-addiction comorbidities, and the difficulty of medication tapering.

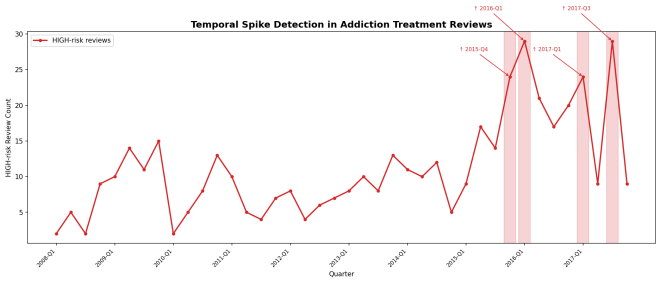

Figure 2: Quarterly HIGH-risk review count with detected spike quarters annotated.

7 Ethical Considerations

No PII. All data is public and self-reported to a public platform. No names, dates of birth, or identifiers are present or inferred.

Population-level only. All outputs describe group-level patterns, not individual risk scores. The system is designed for public health planning, not clinical triage or surveillance of individuals.

Framing. Clusters and summaries are framed around treatment support needs, not criminalization or personal failure. HIGH-risk language detection is presented as a signal for resource allocation, not a patient flag.

Model limitations. LLM labels are probabilistic and may reflect biases in the training data. All LLM outputs are clearly marked as AI-generated and should be reviewed by domain experts before informing policy.

Data access. The raw dataset is publicly available on Kaggle. No scraping, de-anonymization, or secondary collection was performed.

8 Conclusion and Future Work

We demonstrated that unsupervised semantic clustering combined with LLM-based stage labeling can automatically discover interpretable recovery trajectory segments from patient-generated text, without any labeled training data. The pipeline is fully config-driven, reproducible, and outputs analyst-ready markdown summaries and figures.

The survivorship bias finding is the most actionable result: high-risk patients are underrepresented in review data, and this gap should inform any public health study relying on voluntary self-report. Keyword-based detection remains necessary to surface these hidden signals.

Future work includes: (1) integrating Reddit/social media data where high-risk voices are more prevalent; (2) applying larger LLMs (e.g., llama-3.1-70b) for richer cluster characterization; (3) training a lightweight supervised classifier on LLM-labeled reviews to enable real-time scoring; and (4) extending temporal analysis to detect emerging drug trends before they appear in clinical data.

References

[1] F. Gräßer et al., "Aspect-Based Sentiment Analysis of Drug Reviews," *Proc. 2018 Int. Conf. on Digital Health*, 2018.

[2] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," *EMNLP*, 2019.

[3] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction," *arXiv:1802.03426*, 2018.

[4] R. Campello, D. Moulavi, and J. Sander, "Density-Based Clustering Based on Hierarchical Density Estimates," *PAKDD*, 2013.